

EDUCATION

- **Master of Science in Chemical and Biological Engineering**
Gachon University, South Korea *Sept 2025-*
- **Bachelor of Science in Electronics and Communication Engineering**
Khulna University, Khulna, Bangladesh *CGPA: 3.17*
Jan 2016- Jan 2020

PUBLICATIONS

- **Title:** Design of a 6-bit Threshold Inverter Quantization (TIQ) Flash Analog to Digital Converter (ADC)
 - **Summary:** Developed a project, whose objective was to convert analog signals to Digital signals. It was my postgraduate project. Only CMOS is used in this design so that, there will no variation in the fabrication process, and temperature. 0.25 μ m CMOS technology is used and simulated it in siemens tanner eda
 - **DOI:** <https://doi.org/10.48550/arXiv.2501.04627>
- **Title:** Med-2D SegNet: A Light Weight Deep Neural Network for Medical 2D Image Segmentation
 - **Summary:** We propose Med-2D SegNet, an efficient and accurate deep learning model for medical image segmentation. Featuring the compact Med Block, the model achieves state-of-the-art performance with just 2.07M parameters, demonstrating exceptional cross-dataset generalization. Evaluated across benchmark datasets (KVASIR-SEG, PH2, EndoVis, GLAS), Med-2D SegNet excels in both binary and multi-class segmentation tasks. The model strikes a balance between accuracy and efficiency, advancing medical diagnostic tools in resource-constrained environments.
 - **DOI:** <https://doi.org/10.48550/arXiv.2504.14715>
- **Title:** Brain Controlled Wheelchair with Smart Feature
 - **Summary:** It was my undergraduate research project. In this project, I developed an Arduino-based wheelchair, which could be controlled by brain waves using an Android app and a neurosky microwind 2 brainwaves sensor.
 - **DOI:** <https://doi.org/10.48550/arXiv.2501.03371>

SKILLS

- **Languages:** C, Python, Matlab
- **Tools:** Tanner EDA, Hspice, Tensorflow, Pytorch, Keras, Scikit-learn, Opencv, Pandas, FlaskAPI, GIT, Arduino

EXPERIENCE

- **Anwar Technologies** Dhaka, Bangladesh
Junior Machine Learning Engineer *November 2023 - Present*
 - **OCR & Computer Vision Projects:**
 - * Conducted research on OCR to extract table information for supply chain management, including information extraction from various office documentation.
 - * Designed ML systems for deployment, managing API requests from multiple organizations, and implementing automatic data processing and model training.
 - * Implemented OCR solutions for government documentation (TIN, BIN, TAX Return paper) and HR management systems.
 - **An Advanced OCR System for Handwritten Bangla Text Recognition:**
 - * Designed and implemented a state-of-the-art OCR system for accurate recognition of handwritten Bangla text at both sentence and word levels.
 - * Developed a robust OCR model using CRNN architecture with a ResNet34 backbone, combining semi-supervised learning techniques to train on both labeled and unlabeled data, improving recognition accuracy and adaptability.
 - * Utilized synthetic text generation and custom Bangla handwritten fonts (developed in-house) to generate a large, diverse dataset, addressing data scarcity and annotation constraints.
 - * Employed semi-supervised learning methods like self-training and consistency regularization to enhance model generalization and reduce manual annotation efforts.

- * Contributed to advancing OCR technology for under-resourced languages by creating scalable, cost-effective solutions for handwritten Bangla text recognition.

- **GenAI for Stake Enterprise AI (Research Project):**

- * Explore Generative AI (GenAI) with a focus on LLAMA 3.1 for enhancing enterprise AI solutions, including chatbots and natural language processing (NLP) applications.
- * Investigate the use of LLAMA 3.1 for generating synthetic data, optimizing prediction algorithms, and developing conversational AI systems for enterprise environments.
- * Apply findings to create chatbots, streamline text-based interaction systems, and develop OCR solutions for extracting and parsing text in large-scale industrial operations.
- * Implement LLAMA 3.1 for enhancing forecasting accuracy, improving inventory management, and streamlining supply chain processes through advanced AI solutions.
- * Develop AI-powered systems for rap lyric generation and analysis, applying GenAI techniques to creative and industrial text applications.

- **Visie**

Machine Learning Engineer(Part Time)

Dhaka, Bangladesh

March, 2023 - Present

- **Face Recognition System:**

- * Collected data from images, videos, and live stream webcams (frames). Performed noise removal, gamma correction, brightness, and contrast adjustment to improve the quality of frames. Used deep learning (MTCNN) model to detect human faces. Used ArcFace to recognise human faces.

ACHIEVEMENTS AND RESPONSIBILITIES

- **Kaggle Competition (Rohlik Orders Forecasting Challenge)**

- * **Rohlik Orders Forecasting Challenge** — Achieved 89th position among 1017 teams.
- * Developed and fine-tuned forecasting models using machine learning techniques to predict customer order behavior for an online grocery platform.
- * Applied advanced feature engineering, model optimization, and ensemble techniques to improve prediction accuracy in a competitive environment.
- * Gained hands-on experience with real-world forecasting data, enhancing proficiency in data analysis and model evaluation.

- **Champion of inter discipline cricket tournament-2019**

REFERENCES

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